

Continuum Mechanics

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Chapter 05

Balance Laws
Part 3



Entropy Production

The Clausius–Duhem Inequality



We have already proved the Clausius–Duhem inequality in Eulerian form:

$$\rho\dot{\eta} = \frac{\rho r}{\theta} - \nabla^{x} \cdot \left(\frac{\mathbf{q}}{\theta}\right) + \sigma, \qquad \sigma(\mathbf{x}, t) \geq 0, \quad \mathbf{x} \in B_t, \quad t > 0. \qquad (5.22)$$

The book uses a slightly different form. To obtain that form, multiply equation (5.22) by the temperature field θ.

An Equivalent Form of the Clausius–Duhem Inequality

Using the product rule,

$$\begin{aligned}\theta\rho\dot{\eta} &= \rho r - \theta\left\{\frac{1}{\theta}\nabla^{\mathrm{x}}\cdot\mathbf{q} + \mathbf{q}\cdot\nabla^{\mathrm{x}}\left(\frac{1}{\theta}\right)\right\} + \theta\sigma \\ &= \rho r - \nabla^{\mathrm{x}}\cdot\mathbf{q} + \theta\mathbf{q}\cdot\frac{1}{\theta^2}\nabla^{\mathrm{x}}\theta \\ &= \rho r - \nabla^{\mathrm{x}}\cdot\mathbf{q} + \frac{\mathbf{q}}{\theta}\cdot\nabla^{\mathrm{x}}\theta + \theta\sigma. \qquad (5.23)\end{aligned}$$

Here we used

$$\nabla^{\mathrm{x}}\left(\frac{1}{\theta}\right) = -\frac{1}{\theta^2}\nabla^{\mathrm{x}}\theta.$$



Definition (Internal Dissipation Density)

Define the *internal dissipation density* by

$$\delta := \theta\rho\dot{\eta} - \rho r + \nabla^{\mathrm{x}} \cdot \mathbf{q}.$$

Equation (5.23) gives

$$\delta - \frac{\mathbf{q}}{\theta} \cdot \nabla^{\mathrm{x}}\theta = \theta\sigma \geq 0. \tag{5.24}$$

Thus the nonnegative entropy production can be expressed in terms of the internal dissipation and the heat-flux contribution.

Definition (Free Energy Density)

Suppose that ϕ is the internal energy density field per unit mass, η is the entropy density field per unit mass, and θ is the temperature field. The quantity

$$\psi(\mathbf{x}, t) = \phi(\mathbf{x}, t) - \theta(\mathbf{x}, t)\eta(\mathbf{x}, t) \tag{5.25}$$

is called the *free energy density field per unit mass*.

Remark

In the classical theory of homogeneous systems undergoing reversible processes, free energy is the portion of the internal energy available for performing work at constant temperature.



Result 5.12 (Free Energy Balance)

Let $\boldsymbol{\phi} : B \times [0, \infty) \rightarrow \mathbb{E}^3$ be a motion of a continuum body with spatial free energy density $\psi(\mathbf{x}, t)$. Then

$$\rho \dot{\psi} = S : L - \rho \eta \dot{\theta} - \frac{\mathbf{q}}{\theta} \cdot \nabla^{\mathrm{x}} \theta - \theta \sigma.$$



Proof.

From equation (5.25),

$$\begin{aligned}\dot{\psi} &= \dot{\phi} - (\theta\eta)^{\cdot} \\ &= \dot{\phi} - \dot{\theta}\eta - \theta\dot{\eta}.\end{aligned}$$

Therefore,

$$\theta\dot{\eta} = \dot{\phi} - \dot{\theta}\eta - \dot{\psi},$$

and multiplication by ρ gives

$$\theta\rho\dot{\eta} = \rho\dot{\phi} - \rho\eta\dot{\theta} - \rho\dot{\psi}.$$

Proof (Cont.)

Equation (5.23) gives

$$\theta \rho \dot{\eta} = \rho r - \nabla^{\chi} \cdot \mathbf{q} + \frac{\mathbf{q}}{\theta} \cdot \nabla^{\chi} \theta + \theta \sigma.$$

Result 5.10 gives

$$\rho \dot{\phi} = S : L - \nabla^{\chi} \cdot \mathbf{q} + \rho r,$$

Proof (Cont.)

Rearranging the local energy balance gives

$$\rho r - \nabla^{\mathrm{x}} \cdot \mathbf{q} = \rho \dot{\phi} - S : L.$$

Substitution into equation (5.23) yields

$$\theta \rho \dot{\eta} = \rho \dot{\phi} - S : L + \frac{\mathbf{q}}{\theta} \cdot \nabla^{\mathrm{x}} \theta + \theta \sigma.$$



Proof (Cont.)

The two expressions for $\theta\rho\dot{\eta}$ are

$$\theta\rho\dot{\eta} = \rho\dot{\phi} - \rho\eta\dot{\theta} - \rho\dot{\psi},$$
$$\theta\rho\dot{\eta} = \rho\dot{\phi} - S : L + \frac{\mathbf{q}}{\theta} \cdot \nabla^{\mathrm{x}}\theta + \theta\sigma.$$

Equating them and cancelling $\rho\dot{\phi}$ gives

$$-\rho\eta\dot{\theta} - \rho\dot{\psi} = -S : L + \frac{\mathbf{q}}{\theta} \cdot \nabla^{\mathrm{x}}\theta + \theta\sigma.$$

Rearranging,

$$\rho\dot{\psi} = S : L - \rho\eta\dot{\theta} - \frac{\mathbf{q}}{\theta} \cdot \nabla^{\mathrm{x}}\theta - \theta\sigma.$$





Corollary (Reduced Clausius–Duhem Inequality)

Since $\sigma \geq 0$ and $\theta > 0$, Result 5.12 implies

$$\rho\dot{\psi} \leq S : L - \frac{\mathbf{q}}{\theta} \cdot \nabla^{x}\theta - \rho\eta\dot{\theta}.$$

This inequality is called the *reduced Clausius–Duhem inequality*.



Summary of Eulerian Variables

The Twenty-Two Eulerian Unknowns



In the Eulerian description, the unknown fields contribute the following scalar components:

field	number	meaning
$[\boldsymbol{\phi}]_i$	3	motion
$[\mathbf{v}]_i$	3	velocity
ρ	1	mass density
S_{ij}	9	Cauchy stress
θ	1	temperature
$[\mathbf{q}]_i$	3	heat flux
ϕ	1	specific internal energy
η	1	specific entropy

Thus there are

$$3+3+1+9+1+3+1+1=22$$

scalar unknowns to be determined.

Equations Available So Far

The kinematic relation gives three equations:

$$[v_i]_m = \frac{\partial \phi_i}{\partial t}, \qquad i = 1, 2, 3,$$

where $v_i = [\mathbf{v}]_i$ and $\phi_i = [\boldsymbol{\phi}]_i$.

The continuity equation gives one equation:

$$\frac{\partial \rho}{\partial t} + \nabla^{\times} \cdot (\rho \mathbf{v}) = 0.$$

The local balance of linear momentum gives three equations:

$$[\rho \dot{\mathbf{v}}]_i = [\nabla^{\times} \cdot S + \rho \mathbf{b}]_i, \qquad i = 1, 2, 3.$$

Equations Available So Far

The local balance of angular momentum gives three independent equations:

$$S_{ij} = S_{ji}, \qquad 1 \leq i < j \leq 3.$$

The local energy balance gives one equation:

$$\rho\dot{\phi} = S : L - \nabla^{\times} \cdot \mathbf{q} + \rho r.$$

Therefore, we have so far

$$3 + 1 + 3 + 3 + 1 = 11$$

equations for 22 scalar unknowns. Constitutive laws are needed to close the system so that it can be solved.



Lagrangian Local Balance Laws

Result 5.13 (Lagrangian Local Conservation of Mass)

Let $\boldsymbol{\phi} : B \times [0, \infty) \to \mathbb{E}^3$ be a motion of a continuum body with deformation gradient field $F(\mathbf{X}, t)$ and spatial mass density field $\rho(\mathbf{x}, t)$. Let $\rho_0(\mathbf{X})$ be the mass density in the reference configuration B. Then Axiom 5.1, conservation of mass, implies

$$\rho_m(\mathbf{X}, t) \det F(\mathbf{X}, t) = \rho_0(\mathbf{X}), \qquad \mathbf{X} \in B, \quad t \geq 0, \tag{5.18}$$

where $\rho_m(\mathbf{X}, t)$ is the material description of the spatial density $\rho(\mathbf{x}, t)$.

Proof.

Equation (5.18) was proved earlier. □



Definition (5.14 (Piola–Kirchhoff Stress Tensors))

Let $S(\mathbf{x}, t)$ be the Cauchy stress field in B_t associated with a motion $\boldsymbol{\phi}$. The *first Piola–Kirchhoff stress tensor* is

$$P(\mathbf{X}, t) := \det F(\mathbf{X}, t)\, S_m(\mathbf{X}, t) F(\mathbf{X}, t)^{-\mathrm{T}},$$

for every $\mathbf{X} \in B$ and $t \geq 0$.

The *second Piola–Kirchhoff stress tensor* is

$$\boldsymbol{\Sigma}(\mathbf{X}, t) := F(\mathbf{X}, t)^{-1} P(\mathbf{X}, t),$$

for every $\mathbf{X} \in B$ and $t \geq 0$.

Result 5.15 (Lagrangian Local Conservation of Momentum)

Let $\boldsymbol{\phi} : B \times [0, \infty) \to \mathbb{E}^3$ be a motion of a continuum body, and let $\rho_0(\mathbf{X})$ denote the mass density field in the reference configuration B. Then the balance of linear momentum implies

$$\rho_0(\mathbf{X})\ddot{\boldsymbol{\phi}}(\mathbf{X}, t) = \nabla^{\mathrm{X}} \cdot P(\mathbf{X}, t) + \rho_0(\mathbf{X})\mathbf{b}_m(\mathbf{X}, t), \qquad \mathbf{X} \in B, \quad t > 0,$$

where P is the first Piola–Kirchhoff stress tensor and $\mathbf{b}_m$ is the material description of the spatial body-force field $\mathbf{b}$.

Proof.

As stated in Axiom 5.2, the balance law for linear momentum for an arbitrary open set $\Omega_t \subseteq B_t$ is

$$\frac{\mathrm{d}}{\mathrm{d}t}\boldsymbol{\ell}[\Omega_t] = \mathbf{r}[\Omega_t],$$

where the linear momentum is

$$\boldsymbol{\ell}[\Omega_t] = \int_{\Omega_t} \rho(\mathbf{x}, t)\mathbf{v}(\mathbf{x}, t)\ \mathrm{d}V_{\mathrm{x}},$$

and the resultant external force is

$$\mathbf{r}[\Omega_t] = \int_{\partial\Omega_t} S(\mathbf{x}, t)\mathbf{n}(\mathbf{x})\ \mathrm{d}A_{\mathrm{x}} + \int_{\Omega_t} \rho(\mathbf{x}, t)\mathbf{b}(\mathbf{x}, t)\ \mathrm{d}V_{\mathrm{x}}.$$

Proof (Cont.)

Performing a change of variables in $\boldsymbol{\ell}[\Omega_t]$ gives

$$\begin{aligned}
\boldsymbol{\ell}[\Omega_t] &= \int_{\Omega_t} \rho(\mathbf{x}, t)\mathbf{v}(\mathbf{x}, t)\ \mathrm{d}V_{\mathrm{x}} \\
&= \int_{\Omega_0} \rho\big(\boldsymbol{\phi}(\mathbf{X}, t), t\big)\mathbf{v}\big(\boldsymbol{\phi}(\mathbf{X}, t), t\big)\det F(\mathbf{X}, t)\ \mathrm{d}V_{\mathrm{X}} \\
&= \int_{\Omega_0} \rho_m(\mathbf{X}, t)\det F(\mathbf{X}, t)\dot{\boldsymbol{\phi}}(\mathbf{X}, t)\ \mathrm{d}V_{\mathrm{X}} \\
&= \int_{\Omega_0} \rho_0(\mathbf{X})\dot{\boldsymbol{\phi}}(\mathbf{X}, t)\ \mathrm{d}V_{\mathrm{X}},
\end{aligned}$$

where the final equality uses Result 5.13, equation (5.18).

Proof (Cont.)

Performing a change of variables in $\mathbf{r}[\Omega_t]$ gives

$$\mathbf{r}[\Omega_t] = \int_{\partial\Omega_0} \det F(\mathbf{X}, t) S\big(\boldsymbol{\phi}(\mathbf{X}, t), t\big) F(\mathbf{X}, t)^{-\mathsf{T}} \mathbf{N}(\mathbf{X}) \, \mathrm{d}A_{\mathrm{X}} + \int_{\Omega_0} \det F(\mathbf{X}, t) \rho\big(\boldsymbol{\phi}(\mathbf{X}, t), t\big) \mathbf{b}\big(\boldsymbol{\phi}(\mathbf{X}, t), t\big) \, \mathrm{d}V_{\mathrm{X}}.$$

Using Definition 5.14 and Result 5.13,

$$\mathbf{r}[\Omega_t] = \int_{\partial\Omega_0} P(\mathbf{X}, t) \mathbf{N}(\mathbf{X}) \, \mathrm{d}A_{\mathrm{X}} + \int_{\Omega_0} \rho_0(\mathbf{X}) \mathbf{b}_m(\mathbf{X}, t) \, \mathrm{d}V_{\mathrm{X}}.$$

Proof (Cont.)

Since Ω_0 is fixed,

$$\frac{\mathrm{d}}{\mathrm{d}t}\boldsymbol{\ell}[\Omega_t] = \int_{\Omega_0} \rho_0(\mathbf{X})\ddot{\boldsymbol{\phi}}(\mathbf{X}, t)\, \mathrm{d}V_{\mathrm{X}}.$$

Using the Divergence Theorem in the expression for the resultant force,

$$\mathbf{r}[\Omega_t] = \int_{\Omega_0} \left\{\nabla^{\mathrm{X}} \cdot P(\mathbf{X}, t) + \rho_0(\mathbf{X})\mathbf{b}_m(\mathbf{X}, t)\right\} \mathrm{d}V_{\mathrm{X}}.$$

Therefore, the balance of linear momentum gives

$$0 = \int_{\Omega_0} \left\{\rho_0\ddot{\boldsymbol{\phi}} - \nabla^{\mathrm{X}} \cdot P - \rho_0\mathbf{b}_m\right\} \mathrm{d}V_{\mathrm{X}}.$$



Proof (Cont.)

Because the open set $\Omega_0 \subseteq B$ is arbitrary, the Localization Theorem gives

$$\rho_0(\mathbf{X})\ddot{\boldsymbol{\phi}}(\mathbf{X}, t) = \nabla^{\mathrm{X}} \cdot P(\mathbf{X}, t) + \rho_0(\mathbf{X})\mathbf{b}_m(\mathbf{X}, t),$$

for every $\mathbf{X} \in B$ and $t > 0$. □



Result 5.16 (Lagrangian Angular Momentum Balance)

Let $\boldsymbol{\phi} : B \times [0, \infty) \to \mathbb{E}^3$ be a motion of a continuum body with associated deformation gradient $F(\mathbf{X}, t)$. Then the balance of angular momentum requires

$$P(\mathbf{X}, t)F(\mathbf{X}, t)^{\mathsf{T}} = F(\mathbf{X}, t)P(\mathbf{X}, t)^{\mathsf{T}}, \qquad \mathbf{X} \in B, \quad t > 0,$$

or, equivalently,

$$\Sigma(\mathbf{X}, t)^{\mathsf{T}} = \Sigma(\mathbf{X}, t), \qquad \mathbf{X} \in B, \quad t > 0,$$

where P is the first Piola–Kirchhoff stress tensor and Σ is the second Piola–Kirchhoff stress tensor.

Proof.

By Result 5.8, $S_m^{\mathsf{T}} = S_m$. Definition 5.14 gives

$$PF^{\mathsf{T}} = \det(F) S_m, \qquad FP^{\mathsf{T}} = \det(F) S_m^{\mathsf{T}}.$$

Hence

$$PF^{\mathsf{T}} = FP^{\mathsf{T}}.$$

Multiplying this identity on the left by F^{-1} and on the right by $F^{-\mathsf{T}}$ gives

$$F^{-1}P = P^{\mathsf{T}}F^{-\mathsf{T}}.$$

Since $\boldsymbol{\Sigma} = F^{-1}P$ and $\boldsymbol{\Sigma}^{\mathsf{T}} = P^{\mathsf{T}}F^{-\mathsf{T}}$, this is equivalent to

$$\boldsymbol{\Sigma}^{\mathsf{T}} = \boldsymbol{\Sigma}.$$



The First Law of Thermodynamics

Result 5.17 (Lagrangian Net Working)

Let $\boldsymbol{\phi} : B \times [0, \infty) \to \mathbb{E}^3$ be a motion of a continuum body with deformation gradient $F(\mathbf{X}, t)$ and first Piola–Kirchhoff stress tensor $P(\mathbf{X}, t)$. Let $\Omega_t \subseteq B_t$ be arbitrary. Then

$$\frac{\mathrm{d}}{\mathrm{d}t} K[\Omega_t] + \int_{\Omega_0} P(\mathbf{X}, t) : \dot{F}(\mathbf{X}, t) \, \mathrm{d}V_{\mathrm{X}} = \mathcal{P}[\Omega_t], \qquad t > 0.$$

Thus, in view of the definition in equation (5.14),

$$\mathcal{W}[\Omega_t] = \int_{\Omega_0} P(\mathbf{X}, t) : \dot{F}(\mathbf{X}, t) \, \mathrm{d}V_{\mathrm{X}}.$$

Proof.

By definition,

$$K[\Omega_t] = \int_{\Omega_t} \frac{1}{2}\rho(\mathbf{x}, t)|\mathbf{v}(\mathbf{x}, t)|^2 \, \mathrm{d}V_{\mathrm{x}},$$

and

$$\mathcal{P}[\Omega_t] = \int_{\Omega_t} \rho(\mathbf{x}, t)\mathbf{b}(\mathbf{x}, t) \cdot \mathbf{v}(\mathbf{x}, t) \, \mathrm{d}V_{\mathrm{x}} + \int_{\partial\Omega_t} \Big(S(\mathbf{x}, t)^{\mathsf{T}}\mathbf{v}(\mathbf{x}, t)\Big) \cdot \mathbf{n}(\mathbf{x}) \, \mathrm{d}A_{\mathrm{x}}.$$

Here we used

$$(S^{\mathsf{T}}\mathbf{v}) \cdot \mathbf{n} = \mathbf{v} \cdot (S\mathbf{n}).$$

Proof (Cont.)

Performing a change of variables in $K[\Omega_t]$ gives

$$K[\Omega_t] = \int_{\Omega_0} \frac{1}{2}\rho\big(\boldsymbol{\phi}(\mathbf{X}, t), t\big)\,\big|\mathbf{v}\big(\boldsymbol{\phi}(\mathbf{X}, t), t\big)\big|^2 \det F(\mathbf{X}, t)\ \mathrm{d}V_{\mathrm{X}}$$
$$= \int_{\Omega_0} \frac{1}{2}\rho_0(\mathbf{X})|\dot{\boldsymbol{\phi}}(\mathbf{X}, t)|^2\ \mathrm{d}V_{\mathrm{X}}. \tag{5.30}$$

We used

$$\mathbf{v}_m(\mathbf{X}, t) = \dot{\boldsymbol{\phi}}(\mathbf{X}, t)$$

and Result 5.13,

$$\rho_m(\mathbf{X}, t) \det F(\mathbf{X}, t) = \rho_0(\mathbf{X}).$$

Proof (Cont.)

Changing variables in the power gives

$$\begin{aligned}\mathcal{P}[\Omega_t] &= \int_{\Omega_0} \rho_0(\mathbf{X})\mathbf{b}_m(\mathbf{X},t)\cdot\dot{\boldsymbol{\phi}}(\mathbf{X},t)\,\mathrm{d}V_{\mathrm{X}} \\ &\quad + \int_{\partial\Omega_0} \det F(\mathbf{X},t)\mathbf{v}_m(\mathbf{X},t)\cdot S_m(\mathbf{X},t)F(\mathbf{X},t)^{-\mathsf{T}}\mathbf{N}(\mathbf{X})\,\mathrm{d}A_{\mathrm{X}} \\ &= \int_{\Omega_0} \rho_0\mathbf{b}_m\cdot\dot{\boldsymbol{\phi}}\,\mathrm{d}V_{\mathrm{X}} + \int_{\partial\Omega_0}\left(P^{\mathsf{T}}\dot{\boldsymbol{\phi}}\right)\cdot\mathbf{N}\,\mathrm{d}A_{\mathrm{X}}.\end{aligned}$$

The final line uses Definition 5.14 and $\mathbf{v}_m = \dot{\boldsymbol{\phi}}$.

Proof (Cont.)

By the Divergence Theorem and Result 2.11,

$$\begin{aligned}\mathcal{P}[\Omega_t] &= \int_{\Omega_0} \left\{ \rho_0 \mathbf{b}_m \cdot \dot{\boldsymbol{\phi}} + \nabla^{\mathrm{X}} \cdot \left(P^{\mathsf{T}} \dot{\boldsymbol{\phi}} \right) \right\} \mathrm{d}V_{\mathrm{X}} \\ &= \int_{\Omega_0} \left\{ \left(\nabla^{\mathrm{X}} \cdot P + \rho_0 \mathbf{b}_m \right) \cdot \dot{\boldsymbol{\phi}} + P : \nabla^{\mathrm{X}} \dot{\boldsymbol{\phi}} \right\} \mathrm{d}V_{\mathrm{X}}. \end{aligned} \tag{5.31}$$

Indeed, Result 2.11 gives the product rule

$$\nabla^{\mathrm{X}} \cdot (P^{\mathsf{T}} \dot{\boldsymbol{\phi}}) = (\nabla^{\mathrm{X}} \cdot P) \cdot \dot{\boldsymbol{\phi}} + P : \nabla^{\mathrm{X}} \dot{\boldsymbol{\phi}}.$$

Proof (Cont.)

Taking the time derivative of equation (5.30) and using Result 5.15 gives

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}t}K[\Omega_t] &= \int_{\Omega_0} \rho_0 \dot{\boldsymbol{\phi}} \cdot \ddot{\boldsymbol{\phi}} \, \mathrm{d}V_{\mathrm{X}} \\ &= \int_{\Omega_0} \dot{\boldsymbol{\phi}} \cdot \left(\nabla^{\mathrm{X}} \cdot P + \rho_0 \mathbf{b}_m \right) \mathrm{d}V_{\mathrm{X}}.\end{aligned}$$

Also,

$$\nabla^{\mathrm{X}} \dot{\boldsymbol{\phi}}(\mathbf{X}, t) = \dot{F}(\mathbf{X}, t).$$

Subtracting the expression for $\mathrm{d}K/\mathrm{d}t$ from equation (5.31) yields

$$\mathcal{P}[\Omega_t] - \frac{\mathrm{d}}{\mathrm{d}t}K[\Omega_t] = \int_{\Omega_0} P : \dot{F} \, \mathrm{d}V_{\mathrm{X}}.$$

The definition of $\mathcal{W}$ in equation (5.14) completes the proof. □

Result 5.18 (Lagrangian Local First Law)

Let $\boldsymbol{\phi} : B \times [0, \infty) \to \mathbb{E}^3$ be a motion of a continuum body with the material fields

$$\Phi(\mathbf{X}, t) := \phi_m(\mathbf{X}, t) = \phi\big(\boldsymbol{\phi}(\mathbf{X}, t), t\big),$$

$$\mathbf{Q}(\mathbf{X}, t) := \det F(\mathbf{X}, t) F(\mathbf{X}, t)^{-1} \mathbf{q}_m(\mathbf{X}, t),$$

and

$$R(\mathbf{X}, t) := r_m(\mathbf{X}, t) = r\big(\boldsymbol{\phi}(\mathbf{X}, t), t\big).$$

Then the First Law of Thermodynamics implies

$$\rho_0 \dot{\Phi} = P : \dot{F} - \nabla^{\mathrm{X}} \cdot \mathbf{Q} + \rho_0 R, \qquad \mathbf{X} \in B, \quad t > 0.$$



Proof.

The First Law of Thermodynamics states

$$\frac{\mathrm{d}}{\mathrm{d}t} U[\Omega_t] = Q[\Omega_t] + \mathcal{W}[\Omega_t], \tag{5.32}$$

where

$$U[\Omega_t] = \int_{\Omega_t} \rho(\mathbf{x}, t)\phi(\mathbf{x}, t)\ \mathrm{d}V_{\mathrm{x}}$$

is the internal energy,

$$Q[\Omega_t] = \int_{\Omega_t} \rho r\ \mathrm{d}V_{\mathrm{x}} - \int_{\partial\Omega_t} \mathbf{q}\cdot\mathbf{n}\ \mathrm{d}A_{\mathrm{x}}$$

is the net heating, and $\mathcal{W}[\Omega_t]$ is the net working of external forces.

Proof (Cont.)

By a change of variables and Result 5.13,

$$\begin{aligned} U[\Omega_t] &= \int_{\Omega_t} \rho(\mathbf{x}, t)\phi(\mathbf{x}, t)\ \mathrm{d}V_{\mathrm{x}} \\ &= \int_{\Omega_0} \rho_m(\mathbf{X}, t)\Phi(\mathbf{X}, t)\det F(\mathbf{X}, t)\ \mathrm{d}V_{\mathrm{X}} \\ &= \int_{\Omega_0} \rho_0(\mathbf{X})\Phi(\mathbf{X}, t)\ \mathrm{d}V_{\mathrm{X}}. \end{aligned} \tag{5.33}$$

Since Ω_0 and ρ_0 are independent of time,

$$\frac{\mathrm{d}}{\mathrm{d}t} U[\Omega_t] = \int_{\Omega_0} \rho_0(\mathbf{X})\dot{\Phi}(\mathbf{X}, t)\ \mathrm{d}V_{\mathrm{X}}.$$

Proof (Cont.)

Similarly, changing variables in the net heating gives

$$\begin{aligned} Q[\Omega_t] &= \int_{\Omega_0} \rho_0(\mathbf{X}) R(\mathbf{X}, t)\, \mathrm{d}V_{\mathrm{X}} \\ &\quad - \int_{\partial\Omega_0} \det F(\mathbf{X}, t) \mathbf{q}_m(\mathbf{X}, t) \cdot F(\mathbf{X}, t)^{-\mathrm{T}} \mathbf{N}(\mathbf{X})\, \mathrm{d}A_{\mathrm{X}} \\ &= \int_{\Omega_0} \rho_0 R\, \mathrm{d}V_{\mathrm{X}} - \int_{\partial\Omega_0} \mathbf{Q} \cdot \mathbf{N}\, \mathrm{d}A_{\mathrm{X}}. \end{aligned} \tag{5.35}$$

The last equality follows from

$$\begin{aligned} \det(F) \mathbf{q}_m \cdot F^{-\mathrm{T}} \mathbf{N} &= \left(\det(F) F^{-1} \mathbf{q}_m\right) \cdot \mathbf{N} \\ &= \mathbf{Q} \cdot \mathbf{N}. \end{aligned}$$

Proof (Cont.)

Substituting equations (5.33) and (5.35), together with Result 5.17, into equation (5.32) gives

$$\int_{\Omega_0} \rho_0 \dot{\Phi} \, \mathrm{d}V_{\mathrm{X}} = \int_{\Omega_0} \rho_0 R \, \mathrm{d}V_{\mathrm{X}} - \int_{\partial\Omega_0} \mathbf{Q} \cdot \mathbf{N} \, \mathrm{d}A_{\mathrm{X}} + \int_{\Omega_0} P : \dot{F} \, \mathrm{d}V_{\mathrm{X}}.$$

By the Divergence Theorem,

$$\int_{\partial\Omega_0} \mathbf{Q} \cdot \mathbf{N} \, \mathrm{d}A_{\mathrm{X}} = \int_{\Omega_0} \nabla^{\mathrm{X}} \cdot \mathbf{Q} \, \mathrm{d}V_{\mathrm{X}}.$$

Therefore,

$$\int_{\Omega_0} \left\{ \rho_0 \dot{\Phi} - P : \dot{F} + \nabla^{\mathrm{X}} \cdot \mathbf{Q} - \rho_0 R \right\} \mathrm{d}V_{\mathrm{X}} = 0.$$



Proof (Cont.)

Since $\Omega_0 \subseteq B$ is arbitrary, the Localization Theorem yields

$$\rho_0(\mathbf{X})\dot{\Phi}(\mathbf{X},t) = P(\mathbf{X},t) : \dot{F}(\mathbf{X},t) - \nabla^{\mathrm{X}} \cdot \mathbf{Q}(\mathbf{X},t) + \rho_0(\mathbf{X})R(\mathbf{X},t),$$

for every $\mathbf{X} \in B$ and $t > 0$. □